

Technology Stack

Date	11 July 2026
Team ID	
Project Name	EduGenie Google Gemini Powered Learning Assistant
Maximum Marks	2 Marks

Define Your Technology Stack:

Technology Stack Details

S.No	Architecture Component / Layer	Technology Chosen	Justification / Purpose
1	Frontend / Client-Side	HTML5, CSS3, JavaScript	Lightweight, browser-native frontend with no framework dependency. CSS animations provide an interactive and visually engaging student experience. Fully responsive across mobile and desktop.
2	Backend / Server-Side	Python 3.10+ with FastAPI	FastAPI offers high-performance async request handling, automatic API documentation (Swagger), and clean RESTful endpoint structure. Python aligns with the team's core expertise and AI library ecosystem.
3	Database / Data Storage	SQLite (PostgreSQL-ready)	SQLite provides zero-configuration local storage for development and lightweight deployment. Architecture is designed to migrate to PostgreSQL for production-scale usage with minimal code change.
4	Cloud / Hosting / Deployment	Local / GitHub + Docker (planned)	Primary deployment is local (Mac M1 optimised). Docker containerisation planned for cloud hosting. GitHub used for version-controlled deployment pipeline. Optimised for resource-constrained devices.
5	Version Control & CI/CD	Git & GitHub	GitHub provides collaborative version control, branch management, and pull request workflows for the 5-member team. Enables code review, issue tracking, and milestone-based project management.
6	Third-Party APIs / Other Tools	Google Gemini API (GeminiAI)	Google Gemini API powers all core AI features: Q&A, quiz generation, concept explanation, text summarisation, and personalised learning path generation. Chosen for its accuracy, multilingual support, and generous free-tier access.